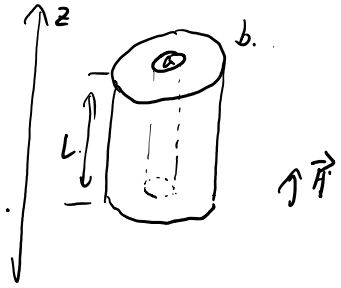
例题

2024年12月3日 8:44

2.17



(1) $\nabla\times\vec{H}=\vec{J}+\frac{\partial\vec{D}}{\partial t}$

$$=\frac{1}{\rho}\begin{vmatrix}\vec{e_\rho} & \vec{e_\varphi} & \vec{e_z}\\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\varphi} & \frac{\partial}{\partial z}\\ H_\rho & \rho H_\varphi & H_z\end{vmatrix}$$

$$=\frac{1}{\rho}\begin{vmatrix}\vec{e_\rho} & \vec{e_\varphi} & \vec{e_z}\\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\varphi} & \frac{\partial}{\partial z}\\ 0 & H_0\cos kz\cos\omega t & 0\end{vmatrix}$$

$$=-\frac{1}{\rho}\cdot\vec{e_\rho}\cdot k\cdot H_0-\sin kz\cos\omega t$$

$$=\vec{e_\rho}\cdot\frac{kH_0}{\rho}\sin kz\cdot\cos\omega t$$

$$=\frac{\partial\vec{D}}{\partial t}$$

$$\Rightarrow \vec{D}=\vec{e_\rho}\frac{k}{\omega}\frac{H_0}{\rho}\sin kz\cdot\sin\omega t\cdot$$

$$\Rightarrow \vec{E}=\vec{e_\rho}\frac{kH_0}{\omega\varepsilon\rho}\sin kz\cdot\sin\omega t\cdot$$

(2) $\frac{\partial\vec{D}}{\partial t}=\vec{e_\rho}\cdot\frac{kH_0}{\rho}\sin kz\cdot\cos\omega t\cdot$

则 $I=\int_s\frac{\partial\vec{D}}{\partial t}ds$

$$=\int_0^{2\pi}\int_0^L\frac{\partial\vec{D}}{\partial t}\rho\cdot d\varphi\cdot dz$$

$$=2\pi\int_0^L\frac{kH_0}{\rho}\cdot\rho\cdot\sin kz\cdot\cos\omega t\,dz$$

$$=2\pi\cos\omega t\int_0^L kH_0\sin kz\cdot dz$$

$$=2\pi\cos\omega t\cdot(-H_0\cos kz)\Big|_0^L$$

$$=2\pi H_0\cos\omega t\,(1-\cos kL)$$

由边界条件，

$$\frac{1}{\rho}\begin{vmatrix}\vec{e_r} & \rho\vec{e_\varphi} & \vec{e_z}\\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\varphi} & \frac{\partial}{\partial z}\\ E_\rho & \rho E_\varphi & E_z\end{vmatrix}=-\frac{\partial\vec{B}}{\partial t}=-\mu\frac{\partial\vec{H}}{\partial t}=\vec{e_\varphi}\frac{\mu H_0\omega}{\rho}\cdot\cos kz\cdot\sin\omega t\cdot$$

$$\Rightarrow \frac{1}{\rho}\cdot\vec{e_\rho}\left(\frac{\partial E_z}{\partial\varphi}-\frac{\partial\rho E_\varphi}{\partial z}\right)+\cdot\vec{e_\varphi}\left(\frac{\partial E_\rho}{\partial z}-\frac{\partial E_z}{\partial\rho}\right)+\frac{1}{\rho}\vec{e_z}\left(\frac{\partial\rho E_\varphi}{\partial\rho}-\frac{\partial E_\rho}{\partial\varphi}\right)$$

$$\Rightarrow \vec{e_\varphi}\cdot\frac{\partial E_\rho}{\partial z}=\vec{e_\varphi}\frac{\mu H_0\omega}{\rho}\cdot\cos kz\cdot\sin\omega t\cdot$$

$$\Rightarrow E=\vec{e_\rho}\cdot\frac{\mu H_0\omega}{\rho\cdot k}\cdot\sin kz\cdot\sin\omega t$$

$$\Rightarrow k=\omega\sqrt{\mu\varepsilon}.$$

$E_{1t}=E_{2t}$，由于导体内 $\sigma=\infty$. $E=0$

$$\Rightarrow \vec{E}\Big|_{z=0}=0.\qquad \vec{E}\Big|_{z=L}=0$$

$$\Rightarrow \sin kL=0\Rightarrow k=\frac{2n\pi}{L}=\omega\sqrt{\mu\varepsilon}.\quad n=1,2,\cdots$$

5.22 $\oint H\cdot d\vec{l} = I$

$\Rightarrow H\cdot 2\pi\rho = \frac{I\cdot\pi\rho^2}{\pi a^2}$ $(0<\rho<a)$

$H = \frac{I\rho}{2\pi a^2}$

$\frac{B}{\mu_1}\cdot\pi\rho + \frac{B}{\mu_2}\cdot\pi\rho = I$.

$\Rightarrow B = \frac{I}{\pi\rho}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right) = \frac{\mu_1\mu_2 I}{(\mu_1+\mu_2)\pi\rho}$.

$$\oint_C \boldsymbol{H}\cdot d\boldsymbol{l} = \int_S\left(\boldsymbol{J}+\frac{\partial \boldsymbol{D}}{\partial t}\right)\cdot d\boldsymbol{S} \qquad \nabla\times\boldsymbol{H} = \boldsymbol{J}+\frac{\partial \boldsymbol{D}}{\partial t}$$

$$\oint_C \boldsymbol{E}\cdot d\boldsymbol{l} = -\frac{d}{dt}\int_S \boldsymbol{B}\cdot d\boldsymbol{S} \qquad \nabla\times\boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\oint_S \boldsymbol{B}\cdot d\boldsymbol{S} = 0 \qquad \nabla\cdot\boldsymbol{B} = 0$$

$$\oint_S \boldsymbol{D}\cdot d\boldsymbol{S} = \int_V \rho dV \qquad \nabla\cdot\boldsymbol{D} = \rho$$

$$\oint_S \vec{J}\cdot d\vec{S} = -\frac{dQ}{dt} = -\frac{d}{dt}\int_V \rho dV \qquad \nabla\cdot\vec{J} = -\frac{\partial\rho}{\partial t}$$

$$\boldsymbol{D} = \varepsilon\boldsymbol{E}$$
$$\boldsymbol{B} = \mu\boldsymbol{H}$$
$$\boldsymbol{J}_c = \gamma\boldsymbol{E}$$

$\Rightarrow W_e = W_{e1} + W_{e2} + W_{e3}$

$W_{e1} = \frac{1}{2}\mu\cdot\int_0^{2\pi}\int_0^a \frac{I^2r^2}{4\pi^2a^4}\, r dr d$

$= \frac{1}{2}\mu\cdot 2\pi\cdot\frac{I^2r^4}{16\pi^2a^4}\Big|_0^a$

$= \mu\pi\cdot\frac{I^2}{16\pi^2} = \frac{\mu_0 I^2}{16\pi}$.

$W_{e2} = \frac{1}{2}\frac{1}{\mu_1}\int_0^{\pi}\int_a^b \frac{\mu_1^2\mu_2^2 I^2}{(\mu_1+\mu_2)^2\pi^2\rho^2}\cdot\rho d\rho d\varphi$

$= \frac{\pi}{2\mu_1}\cdot\frac{\mu_1^2\mu_2^2 I^2}{(\mu_1+\mu_2)^2\pi^2}\cdot\ln\frac{b}{a}$

$= \frac{\mu_1\mu_2^2 I^2}{2(\mu_1+\mu_2)^2\pi}\ln\frac{b}{a}$

所 $W_e = \frac{\mu_0 I^2}{16\pi} + (\mu_1+\mu_2)\cdot\frac{\mu_1\mu_2 I^2}{2(\mu_1+\mu_2)^2\pi}\cdot\ln\frac{b}{a}$

$= \frac{\mu_0 I^2}{16\pi} + \frac{\mu_1\mu_2 I^2}{2(\mu_1+\mu_2)\pi}\ln\frac{b}{a}$.

(2) $W_e = \frac{1}{2}LI^2$

$\Rightarrow L = \frac{2W_e}{I^2} = \frac{\mu_0}{8\pi} + \frac{\mu_1\mu_2}{2(\mu_1+\mu_2)\pi}\ln\frac{b}{a}$.

6.4 (1). $\vec{e}_x E_0\cdot e^{-jkz}\cdot e^{-j\frac{\pi}{2}} + \vec{e}_y\, 3E_0\cdot e^{-jkz}$.

$= E_0\cdot e^{-jkz}[\vec{e}_x\cdot -j + 3\vec{e}_y]$.

(4). $\vec{e}_y\cdot H_0\cdot e^{-jkz\sin\theta}\cdot e^{j\frac{\pi}{2}}$

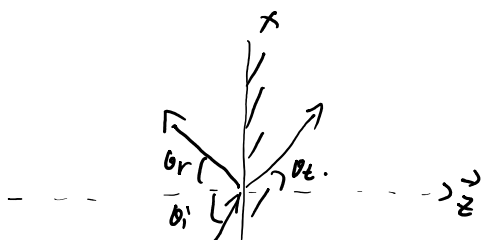
$\vec{e}_y\cdot H_0\cdot\cos(\omega t - kz\sin\theta + \frac{\pi}{2})$.

$= -\vec{e}_y H_0\sin(\omega t - kz\sin\theta)$

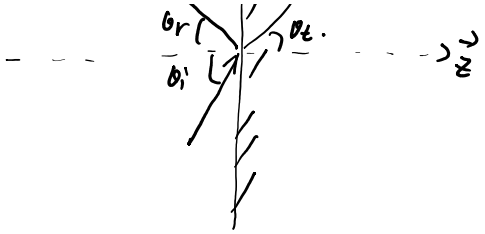
7.40

$e^{-j\vec{k}\cdot\vec{r}} = e^{-j(k_x x + k_y y + k_z z)}$

$\Rightarrow \vec{k} = \frac{\sqrt{3}}{2}\vec{e}_x + \frac{1}{2}\vec{e}_z$



$\Rightarrow \vec{k} = \frac{\sqrt{3}}{2}\vec{e_x} + \frac{1}{2}\vec{e_z}$



则入射角 $\cos\theta_i = \vec{k}\cdot\vec{z} = \frac{1}{2}$

$\Rightarrow \theta_i = 60^\circ$

由反射定律 $\Rightarrow$ $\theta_r = 60^\circ$.

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{k_1}{k_2} = \frac{\sqrt{\mu_1\varepsilon_1}}{\sqrt{\mu_2\varepsilon_2}} = \frac{\sqrt{\varepsilon_{r1}}}{\sqrt{\varepsilon_{r2}}} = \frac{2}{3}.$$

$\Rightarrow \sin\theta_t = \frac{\sqrt{3}}{3}$. $\theta_t = 35.3^\circ$

垂直极化波：(垂直入射面). $\vec{e_y}$.

$$\vec{E}_{\perp i} = \vec{e_y}\cdot e^{-j\vec{k}\vec{r}}$$

$$\eta_1 = \sqrt{\frac{\mu_{r1}}{\varepsilon_{r1}}}\cdot\eta_0 = \frac{1}{2}\eta_0. \quad \eta_2 = \sqrt{\frac{\mu_{r2}}{\varepsilon_{r2}}}\,\eta_0 = \frac{1}{3}\eta_0.$$

$$R_\perp = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} = -0.42.$$

$$T_\perp = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} = 0.58.$$

则 $E_{\perp r} = -0.42\cdot\vec{e_y}\cdot e^{-j\vec{k_r}\cdot\vec{r}}$.

$E_{\perp t} = 0.58\,\vec{e_y}\cdot e^{-j\vec{k_t}\cdot\vec{r}}$.

$\vec{k_r}\cdot\vec{r} = 0(\sqrt{3}x - z)$

$\vec{k_t}\cdot\vec{r} = 0(\sqrt{3}x + \sqrt{3}z)$.

$$\vec{E}_{\parallel i} = (\vec{e_x} - \sqrt{3}\vec{e_z})\cdot e^{-j\vec{k}\cdot\vec{r}}.$$

$$R_\parallel = \frac{\eta_1\cos\theta_i - \eta_2\cos\theta_t}{\eta_1\cos\theta_i + \eta_2\cos\theta_t} = -0.04.$$

$$T_\parallel = \frac{2\eta_2\cos\theta_i}{\eta_1\cos\theta_i + \eta_2\cos\theta_t} = 0.64.$$

$\Rightarrow \vec{E}_{\parallel r} = -0.04(\vec{e_x} - \sqrt{3}\vec{e_z})\cdot e^{-j\vec{k_r}\cdot\vec{r}}$.

$\vec{E}_{\parallel t} = 0.64(\vec{e_x} - \sqrt{3}\vec{e_z})\cdot e^{-j\vec{k_t}\cdot\vec{r}}$.

$\Rightarrow \vec{E}_r = \vec{E}_{r\perp} + \vec{E}_{r\parallel} = (-0.04\vec{e_x} - 0.42\vec{e_y} + 0.04\sqrt{3}\vec{e_z})\cdot e^{-j\sigma(\sqrt{3}x - z)}$.

$\vec{E}_t = \vec{E}_{t\perp} + \vec{E}_{t\parallel}$

7.26. (4). $\vec{E} = \vec{e_x}E_m e^{-jkz}\cdot e^{-j\frac{\pi}{2}} + \vec{e_y}2E_m\cdot e^{-jkz}$.

$\varphi_x - \varphi_y = -\frac{\pi}{2} < 0$. 左旋椭圆极化.

(6). $E_R = \sqrt{5}\vec{e_z}$. $E_I = \vec{e_x} + 2\vec{e_y}$.

$|E_R| = |E_I|$

$E_I \times E_R = (\vec{e_x} + 2\vec{e_y}) \times \vec{e_y}$

$\vec{k} = 2\vec{e_x} - \vec{e_y}$

$E_R\cdot E_I = 0$.

$E_I \times E_R = -\sqrt{5}\vec{e_y} + 2\sqrt{5}\vec{e_x}$.

例：

$$\therefore \quad \oint_S \vec{D}\cdot d\vec{S} = 0$$

所以，根据高斯定律，球壳内表面带电量为$-q$。为保证球壳总电量为Q，须球壳外表面的总电量为Q+q。设内、外球壳上的电荷密度分别为：$-\rho_s, \rho_s'$

则：$\oint_{s_1} -\rho_{s_1} ds_1 = -q, \quad \oint_{s_2} \rho_s' ds_2 = Q+q$

点电荷q在球心处的电位 $\varphi_1 = \dfrac{q}{4\pi\varepsilon_0 r}$

球壳内/外表面上的电荷在球心处产生的电位为：

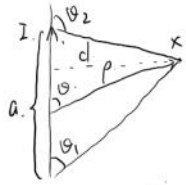
$$\varphi_2 = \oint_{s_1} \frac{-\rho_{s_1} ds_1}{4\pi\varepsilon_0 a} = \frac{-q}{4\pi\varepsilon_0 a}$$

$$\varphi_3 = \oint_{s_3} \frac{\rho_s' ds_2}{4\pi\varepsilon_0 b} = \frac{Q+q}{4\pi\varepsilon_0 b}$$

所以,球心处的总电位为

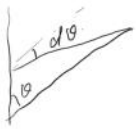
$$\varphi = \varphi_1 + \varphi_2 + \varphi_3 = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{a} + \frac{1}{b}\right) + \frac{Q}{4\pi\varepsilon_0 b}$$

有限长长导线.



$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{I_1 \cdot d\vec{l}_1 \times \vec{e}}{r^2}$$

$$= \frac{\mu_0}{4\pi}\int \frac{I \cdot d\vec{l}}{\left(\frac{d}{\sin\theta}\right)^2} \cdot \sin\theta$$



$$= \frac{\mu_0}{4\pi}\int \frac{I}{d^2}\cdot \sin^3\theta \cdot \frac{1+\tan^2\theta}{\tan^2\theta}\, d\theta \cdot d$$

$$= \frac{\mu_0}{4\pi}\int \frac{I}{d}\cdot \cos^2\theta \, \frac{\sin^2\theta + \cos^2\theta}{\cos^2\theta} \cdot \sin\theta \, d\theta$$

$$= \frac{\mu_0}{4\pi}\int_{\theta_1}^{\theta_2} \frac{I}{d}\cdot \sin\theta \, d\theta$$

$$= \frac{\mu_0 I}{4\pi d}(\cos\theta_1 - \cos\theta_2)$$

$$dl = \frac{d}{\tan\theta} - \frac{d}{\tan(\theta+d\theta)}$$

$$= \frac{d(\tan(\theta+d\theta) - \tan\theta)}{\tan^2\theta}$$

$$= d \cdot \frac{1+\tan^2\theta}{\tan^2\theta}\, d\theta$$

$\theta_1 \approx 0, \ \theta_2 \approx 180$

$$B = \frac{\mu_0 I}{2\pi d}.$$

$$B \cdot 2\pi d = \mu_0 I$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi d}.$$